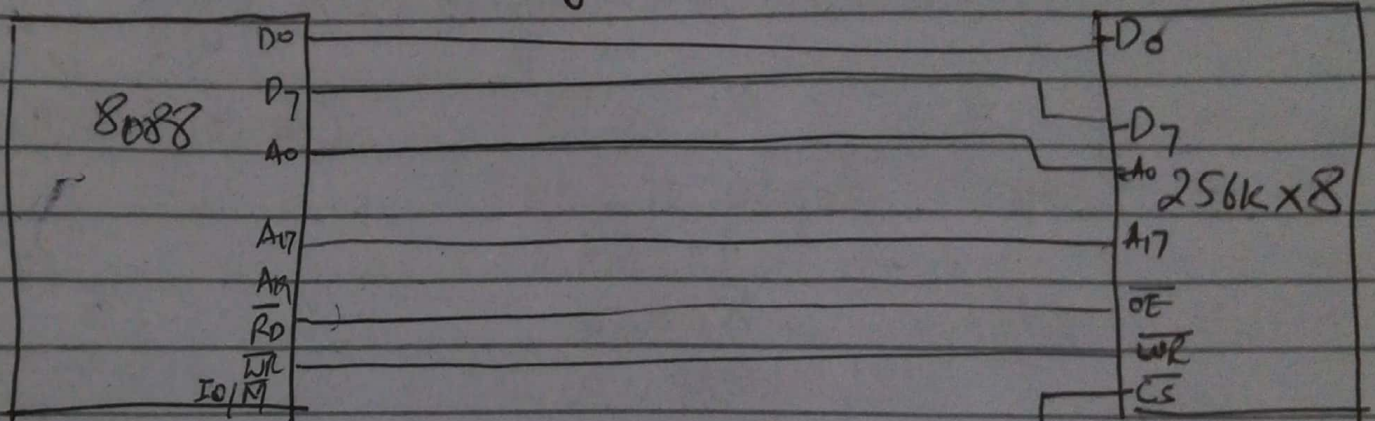


Example

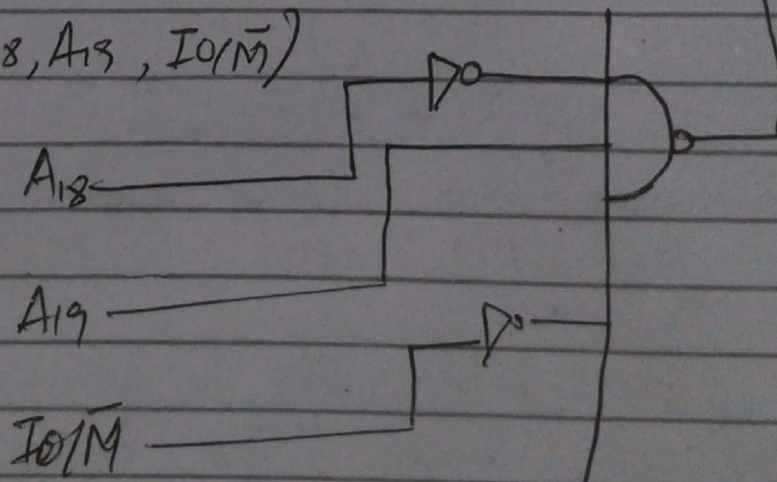
Interface 256K RAM with 8088 Microprocessor
starting address 8000H. Use NAND gate
for Address decoding.

- Tips
- ① Microprocessor
 - ② Memory type + size
 - ③ Starting address
 - ④ Decoding style

Memory Map
Schematic
All other three.



Cleaning pins A18, A19, IO/M



Few values to remember

$$1k = 2^{10} = 400H$$

$$2k = 2^{11} = 800H$$

$$4k = 2^{12} = 1000H$$

$$8k = 2^{13} = 2000H$$

(look yourself)

$$16k = 2^{14} = 4000H$$

$$32k = 2^{15} = 8000H$$

$$64k = 2^{16} = 10000H$$

$$128k = 2^{17} = 20000H$$

$$256k = 2^{18} = 40000H$$

$$512k = 2^{19} = 80000H$$

$$1024k = 2^{20} = 1000000H$$

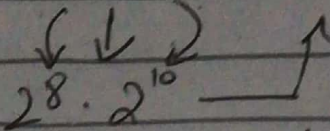
1M

→ Memory available only in these chips

→ What if you need to make a different value like 80k so 64k + 16k chip will be used.

Calculating the address pins

$$256K = 2^8 \cdot 2^{10}$$



so $A_0 - A_{17}$ (Address pins)

Memory Map

starting Address = 80000H
Ending Address

formula Ending Address = Starting Address + size of chips - 1
(As '0' is included)

$$= 80000 + 40000 - 1$$

Hexadecimal Addition

$$\begin{array}{r} 80000 \\ + 40000 \\ \hline C0000 \end{array}$$

$$\begin{array}{r} 8 \\ + 4 \\ \hline 12_{16} \text{ or } E_{16} \end{array}$$

-1

B F F F F

S.A = 80000H = 10000 0000 0000 0000 0000

EA = BFFFFH = 1011 1111 1111 1111 1111

→ In AND gate when any is '0' no one can tell for which I/p the o/p is '0'.

But in case of NAND gate when o/p is '0' both or all I/p's should be '1'. Surely.

NAND Truth Table

A	B	Op
0	0	1
0	1	1
1	0	1
1	1	0

Example (Slide No 17)

16
15
30
3

2K x 8 EPROM, $2^{11} = 2K$

Address bus: $A_{10} - A_0$

Range: 9 pins ($A_{19} - A_{11}$) used for NAND Decoding

Starting Address = FF800H

$$E-A = \text{FF800} \quad \text{S.A} + 2K - 1$$

$$+ \frac{800}{600}$$

$$= \text{FF800H} + 800H - 1$$

$$= \text{FF800H} + 7FFH$$

$$= \text{2FFFFFH}$$

$$\begin{array}{r} 7800 \\ - 1 \\ \hline \end{array}$$

$$\begin{array}{r} 7FF \\ \hline \end{array}$$

$$E-A = \begin{array}{cccccc} 1111 & 1111 & 1111 & 1111 & 1111 \\ & & A_{10} & & A_0 \end{array}$$

$$\begin{array}{r} 7FF \\ \text{FF800} \\ \hline \text{FFFFFH} \end{array}$$

Schematic (In slides No 18)

PFO

Slide 19

64K x 8 SRAM

④

$$64K = 2^6 \cdot 2K = 2^6 \cdot 2^{10} = 2^{16}$$

Address pin $\rightarrow A_0 \rightarrow A_{15}$

Enable Pins $A_{19}-A_{16}$ (4 pins) Connected to NAND

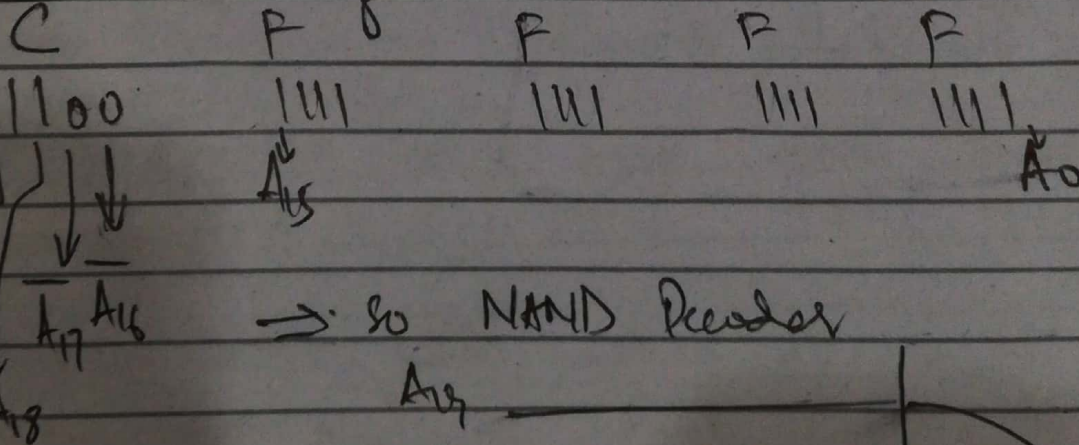
① ^{End} Starting Address $\xrightarrow{SA} E0000H + 64K - 1$
 $= E0000H + 10000H - 1$
 $= F0000H - 1$
 $= EFFFFH$

$\begin{matrix} 5 \\ A_{19} \\ 10000 \end{matrix}$

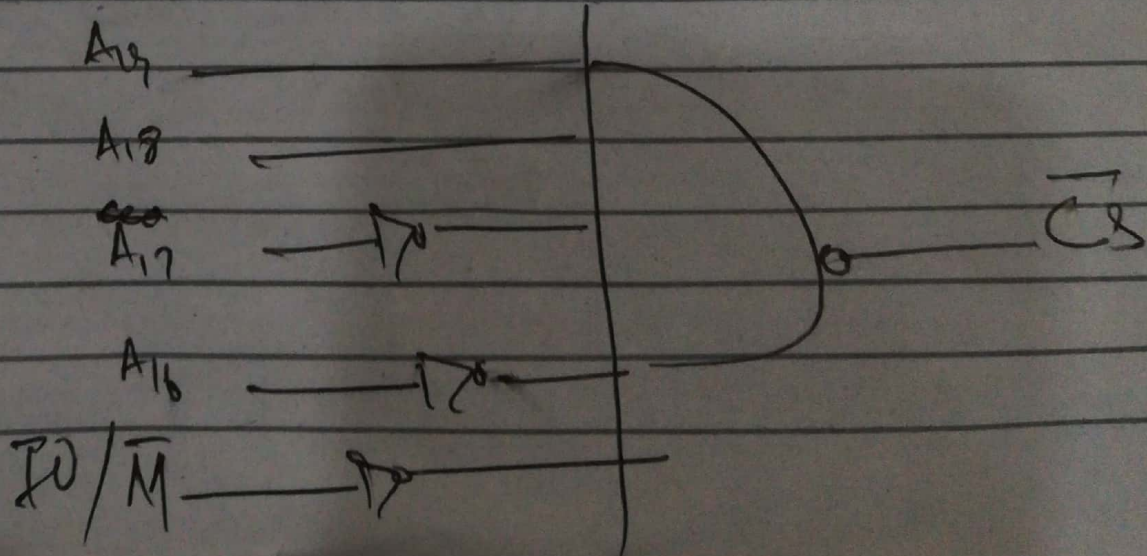
② ^{End} Starting Address $\xrightarrow{SA} C0000H + 64K - 1$
 $= C0000H + 10000 - 1$
 $= D0000H - 1$
 $= CFFFFH$

$\overline{EFFFF}$

So Address Range



$\rightarrow$ So NAND Decoder



slide 22

(5)

Starting Address = 48000H

$$32K \times 8 = 22K = 2^5 \cdot 2K = 2^5 \cdot 2^{10} = 2^{15}$$

Address pins = $A_{14} - A_0$

Address range =

→ Ending Address = S.A + Chip size - 1

$$= 48000H + 8000H - 1$$

$$= 48000H + 7FFFH$$

Binary Pins = 4 F F F F H

↓ In binary

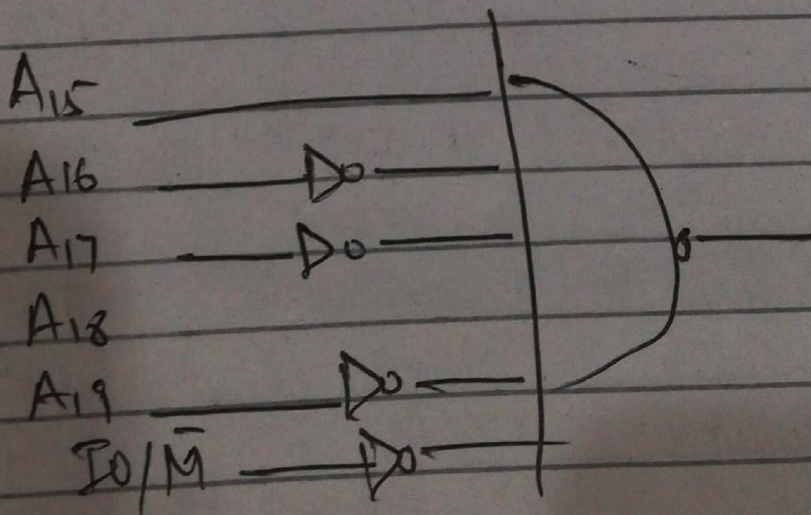
0100 1111
↓
 A_{14}

1111

1111

1111

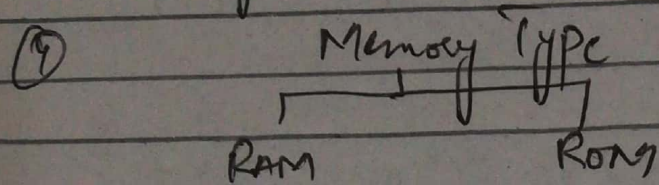
↓
 A_0



8 steps ① Microprocessor 8088
8086

② Decoding technique (a) NAND (b) LINE DECODER

③ Memory Map $\rightarrow$ starting address



S.A = 8E00

Sol: ① Schematic
② Memory Map

16KB (But available chip size is 4KB).

Memory Map

For chip 0 S.A = BE000

$$E.A = BE000_{16} + 1000_{16} - 1$$

$$= BE000_{16} + FFF_{16} = BEFFF$$

For chip 1 S.A = 8F000₁₆

$$E.A = 8F000_{16} + 1000_{16} - 1$$

$$= 8F000_{16} + FFF_{16} = 8FFFF_{16}$$

For chip 2:

$$S.A = 90000_{16}$$

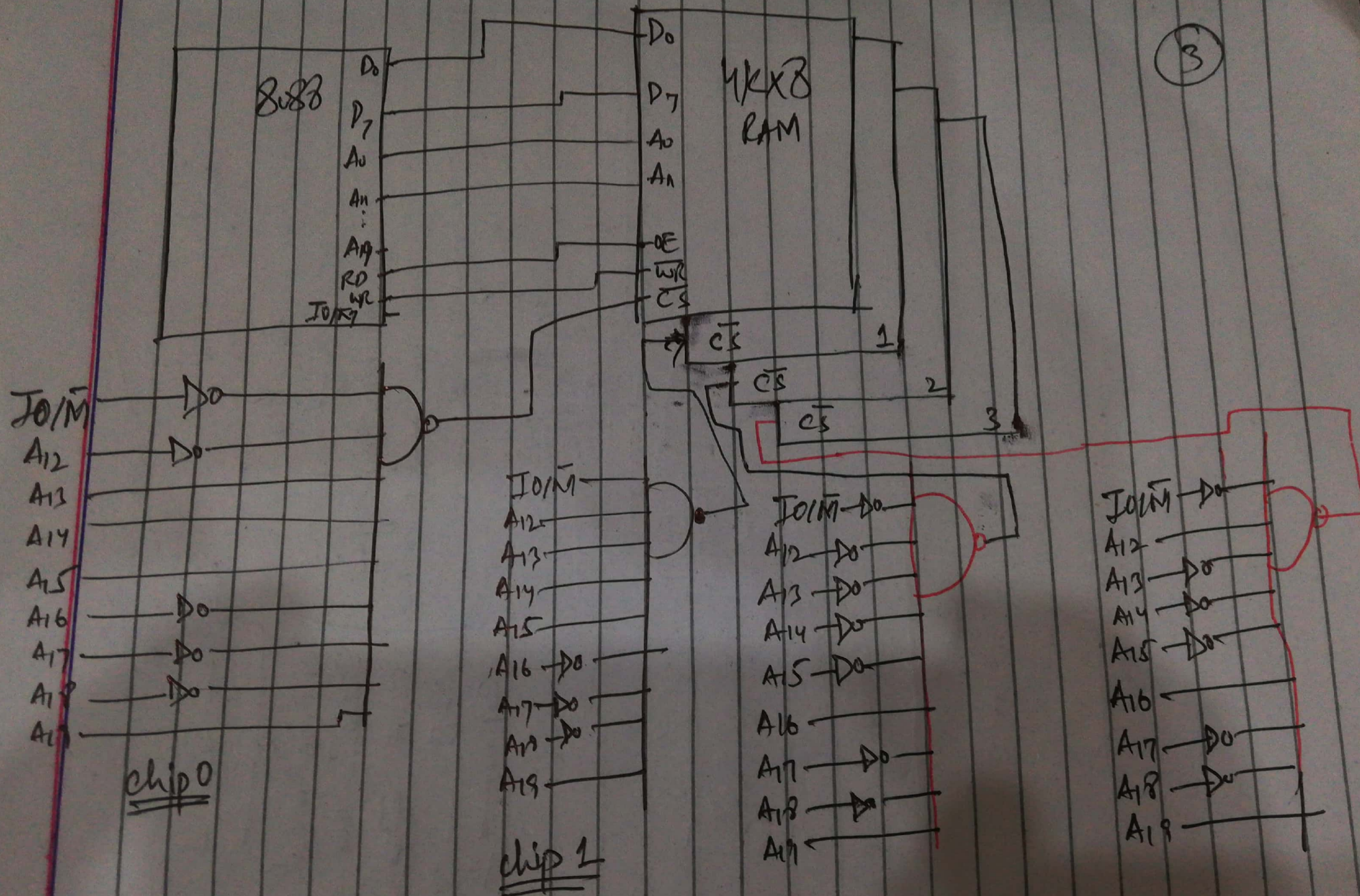
$$E.A = 90FFF_{16}$$

For chip #3

$$S.A = 91000_{16}$$

$$E.A = 91FFF_{16}$$

				Offset Address			(2)
Chip 0	S-A	1000	1110	0000	0000	0000	
	E-A	1000	1110	1111	1111	1111	
				↓ A ₁₁			↓ A ₀
Chip 1	S-A	1000	1111	0000	0000	0000	
	E-A	1000	1111	1111	1111	1111	
Chip 2	S-A	1001	0000	0000	0000	0000	
	E-A	1001	0000	1111	1111	1111	
Chip 3	S-A	1001	0001	0000	0000	0000	
	E-A	1001	0001	1111	1111	1111	
				↓ A ₁₁			↓ A ₀



Example ① Microprocessor = 8088

- ② Type = EPROM 16KB
- ③ Line Decoder
- ④ S.A = 4000h

Sch Schematic
Memory Map

Memory Map
64KB using 16KB = 4 chips
16KB = 4000h $\approx 2^{14}$
-1
43FFF

chip	S.A	E.A
0	40000	43FFF
1	44000	47FFF
2	48000	4BFFF
3	4C000	4FFFF

chip#		A ₁₅					A ₀
		SA	0100	0000	0000	0000	0000
0	EA	0100	0011	1111	1111	1111	1111
chip# 1	SA	0100	0100	0000	0000	0000	0000
	EA	0100	0111	1111	1111	1111	1111
chip# 2	SA	0100	1000	0000	0000	0000	0000
	EA	0100	1011	1111	1111	1111	1111
chip# 3	SA	0100	1100	0000	0000	0000	0000
	EA	0100	1111	1111	1111	1111	1111

For Line decoder

